

Behavior-Aware Unsupervised Intrusion Detection for Zero-Day Attacks Using Host and Network Modeling

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Abstract—This paper presents a data mining approach to network intrusion detection structured around the CRISP-DM methodology. The framework uses behavioral anomaly detection to address zero-day attacks which have no predefined signatures in regular security databases. A dual-layer architecture is adopted that simultaneously analyzes network-level traffic flows using CIC-IDS 2017 [1] and host level system call traces using ADFA-LD [2]. Both models are trained exclusively on benign data 371,621 normal network flows and 833 normal system call traces. Although labeled datasets are used, attack labels are reserved solely for post-hoc evaluation following standard research practice [3] [2] [4]. By modeling normal behavior through association rule mining and n-gram profiling, the framework identifies stable behavioral patterns and flags deviations as potential attacks. The system achieves 92.6% combined recall on known attacks and 54.3% detection on zero-day holdout categories. The results confirm that behavioral anomaly detection is effective against both known and previously unseen attack types.

Index Terms—Intrusion Detection System, Zero-Day Attacks, Behavioral Anomaly Detection, Data Mining, Network Traffic Analysis, System Call Traces, Cybersecurity

I. INTRODUCTION

Traditional Intrusion Detection Systems fail to detect new types of attacks because they are trained only on known threats and patterns. As digital infrastructure grows, new attack types emerge constantly that traditional IDS cannot detect. Zero-Day attack detection solves this by building a behavioral model that learns what normal behavior looks like and flags anything suspicious as abnormal activity, whether the attack is known or completely new. This project builds a dual layer architecture covering both network level and host level attacks. The network model analyzes traffic flows using CIC-IDS 2017 [1] and the host model analyzes system call traces using ADFA-LD [2]. Although both are labeled datasets, labels are never used during training or detection. Both models are trained exclusively on benign data 371,621 normal network flows and 833 normal system call traces. Attack labels are used only after detection for post-hoc evaluation, following standard research practice [3] [2] [4]. The network layer achieves 79.05% recall and 94.33% precision. OR-fusion of

both layers achieves 92.6% combined recall on known attacks and 54.3% detection on zero-day holdout categories.

II. BACKGROUND AND RELATED WORK

Anomaly-based intrusion detection identifies previously unseen attacks by modeling normal behavior rather than relying on predefined attack signatures. Garcia-Teodoro et al. [3] surveyed anomaly detection systems and established that these systems train exclusively on normal behavior. They take a normal day without any attacks to make the model learn what normal looks like. Labeled attack data is used only to measure how well the system detected attacks after the fact, attack labels are never used during training. Early host-based intrusion detection research showed that system call sequences reveal deviations from normal program execution. Creech and Hu [2] built a host-based IDS using normal system call traces only. Attack traces were used only at the end to measure detection rates, not during training. On the network side, association rule mining has been widely used to identify frequent and stable behavioral patterns. Agrawal and Srikant [5] introduced efficient association rule mining and Han et al. [6] improved it through FP-Growth which finds patterns without testing every combination. This project uses FP-Growth directly to mine normal behavioral rules from benign network traffic. Sharafaldin et al. [1] introduced CIC-IDS 2017 which contains benign traffic and a broad range of attacks under realistic conditions. However CIC-IDS captures only network level traffic, missing attacks that manifest through host level system calls. This limitation directly motivates the dual layer design of the present work. Zero-day detection remains difficult because attack behavior is unknown during training and cannot be captured by signatures. Recent studies focused on unsupervised and behavior-aware approaches for zero-day detection [7] [4] [8] [9] and work on generalizable IDS design emphasized models that remain effective under evolving threats [10] [11]. Zoppi et al. [4] demonstrated this approach specifically for zero-day detection, training only on normal data and using attack labels only for evaluation after detection is complete. This project follows this same principle

throughout, both datasets are labeled but labels are used only after detection to measure recall, precision and F1.

III. METHODOLOGY

A. Business Understanding

The digital world is evolving rapidly and new types of attacks are emerging that traditional security systems cannot detect. Traditional IDS systems detect attacks by known patterns and are trained to detect known threats. New threats are seen frequently and many go undetected because traditional IDS are trained to detect only known attacks but fail when new unknown attack types appear. This limitation makes it important to develop a system that focuses on learning normal behavior and detecting attack types, rather than relying only on known attack patterns as traditional IDS do. Zero-day attack detection is a method where the system learns what normal behavior looks like and detects anything which is not normal, whether it is an old attack or a new attack it considers it as abnormal activity. This project develops a dual layer system covering both network level and host level attacks. Although CIC-IDS 2017 and ADFA-LD are labeled datasets, the labels are never used during training or detection. The model is trained exclusively on benign data, learning only what normal behavior looks like. Attack labels are used only for post-hoc evaluation to measure recall, precision and F1, following standard practice in unsupervised anomaly detection research [3] [2] [4]. The target is to achieve 80% or above recall on unseen attack categories. Recall is chosen as the primary metric because missing a real attack is far more dangerous than a false alarm. Accuracy is misleading in this context because network traffic is heavily skewed toward normal data, a model that labels everything as normal would achieve over 99% accuracy while detecting zero attacks.

B. Data Understanding

Two datasets are used, CIC-IDS for network level detection and ADFA-LD for host level detection. CIC-IDS dataset is represented in days format where Monday contains purely normal behavior with no attacks 371,621 benign flows, which is used to train the network model what normal looks like. Other days like Tuesday, Wednesday, Thursday, and Friday each contain different types of network attacks and are used only for post-hoc evaluation. ADFA contains 833 normal system call traces with an average of 370 syscalls each, used to train the host model what normal system behavior looks like. Attack traces in ADFA-LD are used only for post-hoc evaluation. Meterpreter and Web_Shell categories were kept as zero-day holdout, completely excluded from threshold selection to simulate real zero-day conditions. Both models are needed because network model cannot detect host level attacks, some attacks like brute force and XSS manifest through system calls and are invisible to network traffic analysis. The host model covers these missed attack types.

Dataset	Split	Samples	Purpose
CIC-IDS Monday	Training only	371,621 benign flows	Learn normal network behavior
CIC-IDS Tue-Fri	Evaluation only	20 attack categories	Post-hoc evaluation
ADFA-LD Normal	Training only	833 normal traces	Learn normal host behavior
ADFA-LD Attacks	Evaluation only	Known + zero-day holdout	Post-hoc evaluation

TABLE I
DATASETS USED FOR TRAINING AND EVALUATION

C. Data Preparation

CIC-IDS Data Preparation, Monday dataset has 371,621 individual network flows. First IP pair grouping is applied, grouping flows by source IP and destination IP together. This shows the complete activity between two endpoints. When a user logs into a website, the user is the source IP and the website is the destination IP. Grouping by IP pair ensures flows between the same two computers are analyzed together. After IP pair grouping, a sliding window of 20 flows is applied moving forward by 10 flows at a time. Window size of 20 was selected because smaller windows produced too many windows and too many rules, testing showed FPR of nearly 80% which made the model useless. Step size of 10 creates 50% overlap between consecutive windows ensuring no attack activity gets missed at window boundaries. This produces 27,619 windows from 371,621 flows. The original dataset has nearly 80 features but many are zero or null for most flows making them useless, and many are highly correlated with each other making them redundant. 15 features were selected using Pearson correlation to keep only the most informative and independent features. These cover five major behavioral categories packet behavior, flow timing, packet rate, byte rate and volume. These five categories cover all major aspects of network behavior so any attack type will show abnormality in at least one of them. For example a PortScan attack generates abnormal packet rate with low bytes, breaking packet behavior rules. A DDoS attack floods the network showing abnormal byte rate and flow timing. For each window, mean, max and standard deviation are computed per feature giving $15 \times 3 = 45$ values per window. Each value is then converted to LOW, MED, and HIGH using 33rd and 66th percentiles from Monday benign training data. LOW means below normal range, MED means within normal range, HIGH means above normal range. This conversion is needed because FP-Growth works with categories not raw numbers. Finally each window is one hot encoded producing 27,619 windows x 135 items ready for FP-Growth association rule mining. ADFA Data Preparation, this dataset contains 833 normal text trace files. Each file contains a sequence of system call numbers where each represents a command the program sent to the operating system, for example open, read, write, or close. Every time a program runs it generates a trail of system calls. Normal programs leave predictable trails. Attack programs leave unusual trails that were never seen during training. Single syscall numbers make no sense alone, every program uses open and read. By grouping them into sequence we can understand what is actually happening. Bigrams connect two consecutive number, for example 45_102 represents open then read. Trigrams connect three consecutive numbers, for

example 45_102_67 represents open then read then write. Bigrams catch simple two step patterns and trigrams catch more complex three step patterns. Together they give a richer picture of normal behavior. A sliding window of 20 syscalls with step size of 10 is applied to each trace, same reasoning as CIC-IDS, smaller windows produced too much noise. Processing all 833 normal traces produced 29,605 windows. From these windows 12,370 unique n-grams were found 2,116 bigrams and 10,254 trigrams. N-grams appearing below the 25th percentile frequency threshold of 0.000068 are marked as rare 5,123 n-grams representing 41.4% of vocabulary. At test time windows containing high proportions of rare or unseen n-grams receive higher anomaly scores indicating potential attacks.

D. Modeling

FP-Growth is used because it finds patterns from data without needing labels. Unlike other algorithms it finds behavioral relationships among features, showing how features relate to each other which represents normal behavior. Previous algorithms like Apriori tested every possible combination of features which was slow. FP-Growth builds a tree structure and finds patterns without testing every combination making it much faster. Most importantly it produces human readable rules that a security team can understand and verify unlike black box models like neural networks or SVMs. Initially window size 10 and step size 5 produced nearly 7,761 raw rules with a false positive rate of nearly 80% making the model completely useless. To overcome this window size was increased to 20 and step size to 10 which reduced raw rules to 441. These 441 rules were refined through a 7-stage pipeline: redundant rules covered by longer rules were removed reducing from 441 to 431. Rules improving confidence by less than 5% were removed, 431 remained. Mirror rules saying the same thing in opposite directions were removed, 255 remained. Rules were tested across 3 time periods and kept only if valid in at least 2, 255 remained with 223 fully stable. Rules were grouped by features and best rule per group was kept 50 remained. Final 50 rules were selected using composite score. **Composite Score** = $0.4 \times \text{Lift} + 0.3 \times \text{Confidence} + 0.2 \times \text{Support} + 0.1 \times \text{Stability}$

This reduction from 7,761 initial rules to 50 high quality stable rules directly solved the 80% false positive rate problem, demonstrating that rule quality matters more than rule quantity. Each test window is checked against these 50 rules. Anomaly score is calculated as weight of violated rules divided by weight of all applicable rules. A window satisfying 48 of 50 rules scores 0.02 and is considered normal. A window violating 35 rules scores 0.75 and is flagged as an attack. Threshold of 0.10 was selected to prioritize attack detection over false alarms. From ADFA data preparation, 29,605 training windows were produced with 12,370 unique n-grams and 5,123 rare n-grams marked. The ADFA model scores each test window using two components, frequency score and rule score. Frequency score measures how many n-grams in the test window are rare or completely unseen compared to

normal training. During training the model learns which n-grams appear frequently in normal behavior. Normal windows contain common familiar sequences scoring low. Attack windows contains many rare or unseen sequence scoring high.

Formula:

frequency score = number of rare or unseen n-grams / total n-grams in window, capped at 0.80.

Rule score follows the same idea as CIC-IDS but for syscall bigrams. During training FP-Growth found normal bigram patterns, for example 45_102 appearing with 102_67 ninety percent of the time in normal traces. At test time attack windows violate these normal patterns increasing the rule score. Frequency score catches completely new attack sequences while rule score catches subtle pattern violation, together they provide stronger detection than either component alone. Both are combined as:

raw score = $0.6 \times \text{frequency score} + 0.4 \times \text{rule score}$.

Frequency score is given higher weight of 0.6 because completely new attack sequences always show up as unseen n-grams making it more reliable for zero-day detection. The raw score is normalized using:

$$final_score = \sigma \left(\frac{raw_score - \mu}{\sigma_n} \right) \quad (1)$$

where μ and σ_n are computed from 400 normal training traces, ensuring scores are comparable across traces of different lengths. This ensures scores are comparable across traces of different lengths. Threshold of 0.45 was selected by sweeping from 0.45 to 0.95 on known attack categories only. Windows scoring above 0.45 are flagged as attacks. Meterpreter and Web_Shell holdout categories were only scored after threshold selection was complete, a genuine zero-day simulation.

E. Evaluation

Labels are used only after detection is complete to cross verify results, to check how many attacks were correctly identified and how many missed. This model is trained on normal behavior only and does not know what attacks look like. Label dataset is used only to measure recall, precision and F1, following standard research practice [3] [2] [4]. Threshold of 0.10 was selected for CIC-IDS because low threshold makes the model more sensitive, catching more attacks. For the ADFA model, threshold values between 0.45 and 0.95 were swept on known attack categories only, and 0.45 was selected as the best balance between detection and false positives. The ADFA-LD dataset [2] was used only for evaluation after training was complete. During training only 833 normal traces were used to learn normal behavior. Attack traces were never shown to the model during training. After detection was complete, attack labels were used to check how many attacks the model correctly identified. Meterpreter and Web Shell categories were completely excluded from threshold selection because they represent zero-day attacks the model had never seen. These categories were only scored after all decisions were finalized. The network model could not detect these attacks as they operate through system calls not network

packets. After applying the ADFA host model, 54.3% of these zero-day categories were successfully detected, proving that behavioral anomaly detection can identify attacks never seen during training

F. Deployment

Correlation layer observes both models and flags an attack if either one detects suspicious activity. Three strategies were tested OR-Fusion, AND-Fusion, and weighted average with CIC-IDS weight 0.6 and ADFA weight 0.4. AND fusion was rejected because both models must detect the attack, attacks that only appear in one layer get missed completely. OR fusion was selected because either model detecting suspicious activity is enough to trigger an alarm. Final detection rule is flag as attack if CIC-IDS score is above 0.10 OR ADFA score is above 0.45. Before correlation 4 attack types were completely missed by CIC-IDS alone. After OR fusion zero attack types were completely missed. The mathematical benefit of OR fusion is proven by:

$$R_{combined} = R_{CIC} + R_{ADFA} \times (1 - R_{CIC}) \quad (2)$$

Substituting values:

$$R_{combined} = 0.7905 + 0.6488 \times (1 - 0.7905) = 0.926 \quad (3)$$

This shows ADFA contributes an additional 13.6% beyond CIC-IDS alone.

IV. RESULTS

A. Network Layer: CIC-IDS 2017

The network layer analyzes flow-level features from CIC-IDS 2017, capturing network traffic patterns and protocol violations.

TABLE II
NETWORK LAYER (CIC-IDS) PERFORMANCE

Metric	Value	Threshold
Recall	79.05%	0.10
Precision	94.33%	0.10
F1-Score	0.8609	0.10

CIC-IDS excels at volumetric attacks (DDoS, PortScan) but fails on application-layer attacks with minimal network signatures. The network model correctly detected 79.05% of attacks without using any attack data during training, showing that learning normal behavior is enough to catch most attacks.

TABLE III
NETWORK LAYER BLIND SPOTS

Attack Type	Samples	Recall
DoS GoldenEye	768	0.26%
Web Attack - Brute Force	147	4.08%
Web Attack - XSS	60	10.00%
Average Blind Spot	—	4.78%

B. Host Layer: ADFA-LD

The host layer analyzes Linux system call traces, detecting process-level exploitation and behavioral anomalies.

TABLE IV
HOST LAYER (ADFA-LD) PERFORMANCE

Metric	Value	Threshold
Recall	64.88%	0.50
Precision	83.45%	0.50
F1-Score	0.7300	0.50
Zero-Day Recall	54.3%	—

The host model detected 64.88% of known attacks and 54.3% of zero-day attacks it had never seen before, proving that system call behavior can reveal attacks invisible to the network model.

TABLE V
HOST LAYER: SOLVABLE APPLICATION-LAYER ATTACKS

Attack Type	Traces	Recall
Web Attack - Brute Force	338	75.15%
Web Attack - XSS	118	61.86%
Average	—	68.51%

ADFA-LD complements the network layer by detecting application-layer attacks that manifest as abnormal syscall sequences, invisible to network flow analysis.

C. Correlation Layer: OR-Fusion

Before combining both models, 4 attack types were completely missed. After OR-fusion, no attack types were missed across all 20 categories.

TABLE VI
DUAL-LAYER OR-FUSION: APPLICATION-LAYER COVERAGE

Attack	CIC	ADFA	Combined	Gain
Brute Force	4.08%	75.15%	76.16%	+72.08pp
XSS	10.00%	61.86%	65.68%	+55.68pp
Average	7.04%	68.51%	70.92%	+63.88pp

DoS GoldenEye remains at 0.26%, packets are dropped at the network interface and never reach the host process layer. This is an architectural boundary, not a model deficiency.

D. Summary

Network and host layers detect fundamentally different attack classes. CIC-IDS alone completely missed 4 attack types. After OR-fusion, zero attack types were missed, achieving full coverage across all 20 categories. The mathematical proof:

$$R_{combined} = R_{CIC} + R_{ADFA}(1 - R_{CIC}) \quad (4)$$

$$= 0.7905 + 0.6488(1 - 0.7905) = 0.926 \quad (5)$$

This confirms ADFA contributes an additional 13.6 percentage points beyond CIC-IDS alone.

V. ANALYSIS AND DISCUSSION

CIC-IDS model catches DDoS easily because packet rate is extremely high, byte rate is extremely high, and flow duration is very short, making the traffic clearly abnormal in network flows. The very short inter-arrival time between packets makes it even more suspicious. Brute force attacks are missed by CIC-IDS because packet rate is low, byte rate is low, and flow duration is normal, making it look like regular network behavior. CIC-IDS cannot distinguish this from normal traffic. Similarly XSS attacks inject malicious scripts through web requests, network traffic looks normal but the host process executes unusual syscall sequences, invisible to CIC-IDS but detectable by ADFA. ADFA catches brute force because every login attempt triggers authentication system calls. Thousands of authentication syscalls happen very fast, while a normal user only authenticates once or twice. This unusual syscall sequence is flagged by ADFA as abnormal. DoS GoldenEye remains at 0.26% detection because packets are controlled at network interface level and never reach host process layer, this is an architectural boundary not a model deficiency. These two layers complement each other, attacks missed by one layer are covered by the other. This is proven by results, CIC-IDS alone missed 4 attack types completely. After adding ADFA through OR fusion, zero attack types were missed, achieving full coverage across all 20 attack categories.

VI. LIMITATIONS

The 54.3% zero-day recall, while demonstrating genuine detection capability on unseen attacks, indicates room for improvement. Deep learning approaches such as autoencoders may capture more complex behavioral patterns in future work.

- 1) **Dataset Characteristics:** CIC-IDS 2017 [1] contains labeled attacks, limiting true zero-day validation. Synthetic zero-day tests were created (unseen patterns), which serve as a proxy.
- 2) **Feature Representation:** Sliding window aggregation (20-flow/syscall chunks) may get ambiguous for rapid attack sequences. Smaller windows increase sensitivity but reduce temporal context.
- 3) **Class Imbalance:** Network traffic is heavily skewed toward benign (>99%). Stratified split and balanced class weights mitigate but do not eliminate bias.
- 4) **Threshold Selection:** Anomaly threshold (0.10) is chosen empirically; security teams must tune it based on operational requirements.
- 5) **Real-World Deployment Gaps:** This evaluation assumes a controlled lab environment. Production systems face encrypted traffic (TLS), encrypted syscalls, and adaptive adversaries.
- 6) **Explainability Depth:** Future work includes deep learning for encrypted traffic [8] and incremental learning against concept drift.
- 7) **Dataset Age:** CIC-IDS 2017 [1] is nearly 9 years old and ADFA-LD [2] is nearly 13 years old. Modern attack techniques have evolved significantly since these

datasets were created. Behavioral rules learned from these datasets may not fully represent current attack patterns, potentially reducing effectiveness against modern threats.

VII. CONCLUSION

This paper demonstrates that behavioral anomaly detection following the CRISP-DM methodology effectively identifies network intrusions, including zero-day attacks unseen during training. Results support deployment of machine learning based IDS in real-world security operations as a powerful complement and viable alternative to signature-based systems. The 92.6% combined recall on known attacks and 54.3% zero-day recall establishes practical defense capabilities that address critical gaps in traditional pattern matching approaches. Key findings validate the core research hypothesis, independent network and host layers achieve comprehensive detection through simultaneous specialization, network layer achieving 79.05% recall independently. The CRISP-DM framework structured the project into six explicit phases, from business understanding through deployment, ensuring each design decision is explicitly documented with parameters, making the methodology transparent and independently verifiable. Architectural boundaries like DoS GoldenEye's 0.26% detection rate reflect network-only phenomena, representing design features rather than system deficiencies. Organizations implementing this dual layer approach gain measurable zero-day detection strength, positioning behavioral IDS as critical infrastructure for modern security operations facing advanced persistent threats that evade signature-based detection. Future work should address several critical areas, deep learning architectures could improve the current 54.3% zero-day recall using auto encoders and 1D-CNNs on encrypted traffic patterns. Incremental learning methods for updating behavioral baselines as network conditions evolve, with concept drift detection to identify when training assumptions become invalid. Encrypted traffic analysis for TLS/HTTPS streams using flow statistics, with careful handling of legitimate services like VPNs and backups to minimize false positives.

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